

Richard O’Shaughnessy

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RESEARCH INTERESTS

Gravitational wave astrophysics
Strong field gravity, numerically and analytically
Theoretical physics

EDUCATION

Ph. D. in Physics	Advisor: Kip Thorne	California Institute of Technology	1996-2003
B.A. in Astronomy	<i>summa cum laude</i>	Cornell University	1992-1996

POSITIONS

Rochester Institute of Technology	Associate Professor	2020-
-	Assistant Professor	2014-2020
University of Wisconsin-Milwaukee	Research Associate	2010-2014
Bradley Foundation Fellowship (2011-12)		
Penn State University	Research Associate	2007-2010
Northwestern University	Postdoctoral Fellow	2003-2007
California Institute of Technology	Graduate student and Teaching assistant	1996-2002
Cornell University	Relativity visualization REU group	1995-1996
PIs: Profs. Shapiro and Teukolsky		

TEACHING AND MENTORING

Postdoc projects

- Z. Zhu, RIT (PRD 113 3037+...)
Inference of the nuclear EOS from binary NS observations
- Atul Kedia, RIT (arxiv:24051.7326, PRR 5 013168 2023, GW230529 + ...)
Multimessenger interpretation of kilonova and the nuclear EOS
- E. Holmbeck, RIT (arxiv:2110.06432 + ...)
Multimessenger Observational constraints on r-process nucleosynthesis
- A. Williamson, RIT (PRD 96 404 2017 + ...)
Systematic errors in measuring parameters of precessing BH binaries
- A. Lundgren, Syracuse University and AEI-Hannover, (PRD 86 4020, 2013; arXiv:1304.3332; ...)
Analytic waveforms for precessing BH-NS binaries
- L. Pekowsky and J. Healy (both with D. Shoemaker), Georgia Tech (PRD 88 4040, 2013+ ...)
Interpreting simulations of BH-BH binaries
- E. Ochsner, U. Wisconsin-Milwaukee (PRD 86 4037, 2012)
Analytic methods to reconstruct binary dynamics from outgoing radiation

Graduate PhD student projects

- K. Daningburg, RIT MM (PRD 106 4020 2022)
Improving Neutron Scattering and Fission Modeling with Modern Machine Learning

- M. Zeeshan (Fullbright), RIT ASTP (2607.15431+...) *Population Inference of Compact Binaries with Eccentricity*
- K. Wagner RIT ASTP *Identifying signatures of eccentric binary black hole orbits with GW*
- K. Nathaniel RIT ASTP *Impact of stars on the AGN disk formation channel*
- A. Yelikar, RIT ASTP (PRD 110 4024, 2024; ...) *Quantifying and propagating model sytematics for compact binary mergers*
- M. Ristic, RIT ASTP (PRR 6 3078, 2024; PRR 5 3106, 2023; ...) *Surrogate models for and inference with kilonova light curves*
- V. Delfavero, RIT ASTP (arXiv:2107.13082+...) *Constraining isolated binary evolution with gravitational waves*
- Daniel Wysocki, RIT ASTP (arXiv:1709.01943+...) *Inferring the distribution of merging binary BH masses and spins*
- Jacob Lange, RIT *Parameter estimation via numerical relativity templates* (arxiv:1606.01262+...) *Parameter estimation via iterative inference (RIFT)* (arxiv:1805.10457+...) 2021 RIT PhD Dissertation award
- P. Nepal, U. Wisconsin-Milwaukee (arxiv:1509.06581) *Semianalytic Fisher matrix for precessing BH-NS binaries*
- H. Qi (with P. Brady), U. Wisconsin-Milwaukee (PRD 103 084006, 2021) *Probing strong-field gravity with galactic center orbits*
- B. Farr (with E. Ochsner), Northwestern (PRD 89 102005, 2014 + ...) *Parameter estimation of black hole-neutron star binaries*
- L. London (with D. Shoemaker, L. Pekowsky, J. Healy), Georgia Tech (PRD 87 4038, 2013+...) *Investigating precessing binary black hole simulations*
- H.S. Cho (with CH Lee, C. Kim, E. Ochsner), KISTI and UWM (PRD 87 2400, 2013+...) *Impact of amplitude corrections and spin on parameter estimation for BH-NS binaries in LIGO*
- D. Clausen (with R. Wade, R. Kopparapu), Pennsylvania State University (ApJ 746 186, 2012) *Population synthesis of hot subdwarf binaries: hidden populations and new constraints*

Undergraduate and MS research projects

- Noah Manning (2023-present), RIT BS/MS: Systematics in GW parameter inference
- M. Ebiri (2024-2026), RIT SOPA MS, A HyperPipeline for Joint Inference of the Equation of State and Population of Binary Neutron Stars
- A. Vilkhya (2022-2024), RIT ASTP MS, Efficiently constraining the nuclear equation of state with RIFT (arXiv:2407.15753)
- Manan Kapoor (2022-2023), RIT MM MS, Eccentric binary black holes near SMBH
- A. Jan, RIT SOPA MS (PRD 102 4069 2020) *Implications of model sytematics for compact binary mergers with black holes*
- Elizabeth Champion (2018–2021), RIT BS: Rapid Monte Carlo integration for GW parameter inference
- Vera Delfavero (2018–2019), RIT BS/MS: Gaussian process interpolation for GW parameter inference
- Matteo Breschi, Pisa MS, *Inspiral-merger consistency tests of general relativity with higher modes in LIGO's O2 run* (...)
- Monica Rizzo (2015–2018), RIT BS: Estimating the tidal deformability of neutron stars (GW170817)
- Thomas Kilmer (2016–2017), RIT BS: Subtracting astrophysical stochastic GW foregrounds

- D. Trifiro, Pisa MS, *Parameter estimation with post-Newtonian resonances* (PRD 93 044071, 2016 +...)
- Jackson Henry (2014–), RIT: Higher harmonics and parameter estimation of binary black holes
- Brandon Miller (2014-2015), RIT: Fast versus accurate inference of inspiral (PRD 92 4056, 2015)
- D. Gerosa (with E. Berti, M. Kesden, U. Sperhake), U. Miss. (PRD 87 4028, 2013 +...) *Post-Newtonian spin resonances enable unexpected compact binary astrophysics*
- Z. Meeks (2011), Georgia Tech BS: Orientation-dependent emission from mergers (PRD 85,084003)
- C. Schmidt (2007-2008), Penn State BS: Metadata for numerical relativity waveforms.
- E. Damashek (2007), high school: Strong gravitational lensing close to binary black holes
- A. Saleem (2005-2006), Northwestern BS: Classify progenitors of merging compact binaries.
- J. Kaplan (2004), Northwestern BS: Spinup of black holes in binaries (ApJ 632 1035, 2005)
- R. O’Leary (2004-2006), Northwestern BS: Clusters (PRD 76 061504; ApJ 637 937)
- D. Jones (2003-2004), Northwestern BS: Prototype search code for gravitational waves.

Summer school lectures

- Caltech Gravitational Wave Astrophysics Summer school (2013)

Structured classroom

- *Professor* (RIT)
Math 182, Project-based calculus II (Fall 2014, Fall 2020)
Statistics 435, Statistics of linear models (Spring 2015)
Math 251, Probability and Statistics I (Fall 2015-2019, 2021, 2024; Spring 2018, 2022, 2024-5)
Math 252, Probability and Statistics II (Spring 2017)
Astro 611, Statistical methods for astrophysics (Spring 2016)
Astro 831, Stellar evolution and environments (Spring 2018)
Astro 609, Fundamentals of Astrophysics II (Spring 2019-2021 with A. Robinson)
Astro 619, Fundamentals of Theoretical Astrophysics 2 (Fall 2023, 2024)
- *Instructor* (UWM)
Astronomy 103, Survey of Astronomy (Spring 2012)

PROFESSIONAL AFFILIATIONS AND SERVICE

- Member of the LIGO Scientific Collaboration (2000–);
Co-chair of the Publications and Presentations Committee (2018–)
- Served as referee for Nature; Physical Review Letters; the Monthly Notices of the Royal Astronomical Society; the Astrophysical Journal (regular and letters); Astronomy and Astrophysics; Classical and Quantum Gravity; and the New Journal of Physics.
- Proposal review for NSF and NWO (Netherlands).
- Member of the American Physical Society (APS), the American Astronomical Society (AAS), and the International Astronomical Union (IAU)
- Midwest Relativity Meeting organizing committee (2013)

RESEARCH PUBLICATIONS

Selected recent small collaboration results

1. Z. Zhu and **R. O'Shaughnessy**.
Constraining the Equation of State of Neutron Stars with third-generation Gravitational Wave detectors.
Available as LIGO-P2600366, July 2026.
2. M. Quazalbash, S. Mehra, M. Zeeshan, and **R. O'Shaughnessy**.
Narrow Population Inference Enhanced by Analytical Likelihood Models.
Available as LIGO P2500708, June 2026. (link).
3. M. Zeeshan and **R. O'Shaughnessy**.
Uncovering Hierarchical Sub-Population of Binary Black Holes.
Available as LIGO-P26000337 at dcc.ligo.org, July 2026.
4. Muhammad Zeeshan, Richard O'Shaughnessy, Natalie Malagon, and Katelyn Wagner.
Assessing the waveform systematics from parameter estimation to population inference with eccentricity.
5. Muhammad Zeeshan, Richard O'Shaughnessy, and Natalie Malagon.
Population Properties of Binary Black Holes with Eccentricity.
arXiv e-prints, art. arXiv:2602.11030, February 2026.
6. Aasim Jan, Sophia Nicolella, Deirdre Shoemaker, and Richard O'Shaughnessy.
Measuring eccentricity and addressing waveform systematics in GW231123.
Phys. Rev. D, 113(8):084052, April 2026a.
7. Mahmudul Hasan Anik, Andrew W. Steiner, and Richard O'Shaughnessy.
Inference of Neutron Star Mass Distributions and the Equation of State from Multi-messenger Observations.
arXiv e-prints, art. arXiv:2512.12130, December 2025.
8. M. Qazalbash, M. Zeeshan, and **R. O'Shaughnessy**.
Implementation to identify the properties of multiple populations of gravitational wave sources.
Phys. Rev. D, 113(10):103003, May 2026.
9. G. Gayathri, V. Iorio, H. Tagawa, Daniel Wysocki, J. Anglin, I. Bartos, S. Bhaumik, Z. Haiman, M. Mapelli, **R. O'Shaughnessy**, and L. Xue.
Reconstructing the origin of black hole mergers using sparse astrophysical models.
Submitted to Astronomy and Astrophysics, available as *arXiv:2509.09647*, art. arXiv:2509.09647, September 2025. (link).
10. V. Delfavero, S. Ray, H. E. Cook, K. Nathaniel, B. McKernan, K. E. S. Ford, J. Postiglione, E. McPike, and **R. O'Shaughnessy**.
Prospects for the formation of GW231123 from the AGN channel.
Submitted to PRL, art. arXiv:2508.13412, August 2025a.
11. Aasim Jan, Bing-Jyun Tsao, Richard O'Shaughnessy, Deirdre Shoemaker, and Pablo Laguna.
GW200105: A detailed study of eccentricity in the neutron star-black hole binary.
Phys. Rev. D, 113(2):024018, January 2026b.
12. Z. Zhu and **R. O'Shaughnessy**.
Limitations in constraining neutron star radii and nuclear properties from inspiral gravitational wave detections.
Submitted to PRL, available as *arxiv:2508.11875*, August 2025. (link).
13. Chad Henshaw, Jacob Lange, Peter Lott, Richard O'Shaughnessy, and Laura Cadonati.
Parameter estimation of gravitational waves from hyperbolic black hole encounters.
Classical and Quantum Gravity, 42(24):24LT01, December 2025.

14. Patricia McMillin, Katelyn J. Wagner, Giuseppe Ficarra, Carlos O. Lousto, and Richard O’Shaughnessy. Parameter Estimation for GW200208_22 with Targeted Eccentric Numerical-relativity Simulations. *arXiv e-prints*, art. arXiv:2507.22862, July 2025.
15. K. Wagner, **R. O’Shaughnessy**, A. Yelikar, N. Manning, D. Fernando, J. Lange, V. Tiwari, and A.H. Fernando. Narrowing RIFT: Focused simulation-based-inference for interpreting exceptional GW sources. *Submitted to PRD; available as arxiv:2505.11655*, May 2025. (link).
16. Marko Ristic, Richard O’Shaughnessy, K. Wagner, Christopher J. Fontes, Christopher L. Fryer, Oleg Korobkin, M. Mumpower, and Ryan T. Wollaeger. Joint Electromagnetic and Gravitational Wave Inference of Binary Neutron Star Merger GW170817 Using Forward-Modeling Ejecta Predictions. *Submitted to PRR; available as arxiv:2503.12320*, March 2025. (link).
17. Krystal Ruiz-Rocha, Anjali B. Yelikar, Jacob Lange, William Gabella, Robert A. Weller, Richard O’Shaughnessy, Kelly Holley-Bockelmann, and Karan Jani. Properties of “Lite” Intermediate-mass Black Hole Candidates in LIGO-Virgo’s Third Observing Run. *Astroph. J. Letters*, 985(2):L37, June 2025.
18. D. Fernando, Richard O’Shaughnessy, and Daniel Williams. Efficient reanalysis of events from GWTC-3. *Phys. Rev. D*, 112(8):084004, October 2025.
19. Barry McKernan, K. E. Saavik Ford, Harrison E. Cook, Vera Delfavero, Emily McPike, Kaila Nathaniel, Jake Postiglione, Shawn Ray, and Richard O’Shaughnessy. McFACTS I: Testing the LVK AGN Channel with Monte Carlo for AGN Channel Testing and Simulation (McFACTS). *Astroph. J.*, 990(2):217, September 2025.
20. Harrison E. Cook, Barry McKernan, K. E. Saavik Ford, Vera Delfavero, Kaila Nathaniel, Jake Postiglione, Shawn Ray, Emily J. McPike, and Richard O’Shaughnessy. McFACTS. II. Mass Ratio-Effective Spin Relationship of Black Hole Mergers in the Active Galactic Nucleus Channel. *Astroph. J.*, 993(2):163, November 2025.
21. Vera Delfavero, K. E. Saavik Ford, Barry McKernan, Harrison E. Cook, Kaila Nathaniel, Jake Postiglione, Shawn Ray, Emily McPike, and Richard O’Shaughnessy. McFACTS III: Compact Binary Mergers from Active Galactic Nucleus Disks over an Entire Synthetic Universe. *Astroph. J.*, 989(1):67, August 2025b.
22. Emily McPike, Rosalba Perna, K. E. Saavik Ford, Barry McKernan, Vera Delfavero, Miranda McCarthy, Kaila Nathaniel, Jake Postiglione, Nicolas Posner, Varun Pritmani, Shawn Ray, and Richard O’Shaughnessy. McFACTS IV: Electromagnetic Counterparts to AGN Disk Embedded Binary Black Hole Mergers. *arXiv e-prints*, art. arXiv:2602.04135, February 2026.
23. Aasim Jan, Richard O’Shaughnessy, Deirdre Shoemaker, and Jacob Lange. Adapting a novel framework for rapid inference of massive black hole binaries for LISA. *Phys. Rev. D*, 111(6):064079, March 2025.
24. Vera Delfavero, K. Breivik, **R. O’Shaughnessy**, and J.G. Baker. Recovering Injected Astrophysics from the LISA Galactic Binaries. *Astroph. J.*, 981:66, February 2025c.
25. Anjali B. Yelikar, Richard O’Shaughnessy, Daniel Wysocki, and Leslie Wade. Effects of waveform systematics on inferences of neutron star population properties and the nuclear equation of state. *Submitted to PRD; available as arxiv:2410.14674*, art. arXiv:2410.14674, October 2024a.

26. Askold Vilkhia, Anjali Yelikar, Richard O’Shaughnessy, and Jocelyn Read.
Inference on neutron star parameters and the nuclear equation of state with RIFT, using prior EOS information.
arXiv e-prints, art. arXiv:2407.15753, July 2024.
27. K. Daningburg, A. Lovell, and **R. O’Shaughnessy**.
A novel emulator for fission event generators.
Submitted to PRC, available as arxiv:2409.16406, September 2024.
28. A. B. Yelikar, **R. O’Shaughnessy**, J. Lange, and A. Z. Jan.
Waveform systematics in gravitational-wave inference of signals from binary neutron star merger models incorporating higher-order modes information.
Phys. Rev. D, 110(6):064024, September 2024b.
29. M. Zeeshan and **R. O’Shaughnessy**.
Eccentricity matters: Impact of eccentricity on inferred binary black hole populations.
Phys. Rev. D, 110(6):063009, September 2024.
30. A. Kedia, **R. O’Shaughnessy**, L. Wade, and A. Yelikar.
Exploring hidden priors when interpreting gravitational wave and electromagnetic probes of the nuclear equation of state.
Submitted to PRD, available as arxiv:2405.17326, May 2024. (link).
31. Katelyn J. Wagner and **R. O’Shaughnessy**.
Parameter estimation for low-mass eccentric black hole binaries.
Phys. Rev. D, 110(12):124024, December 2024.
32. Yinglei Peng, Marko Ristić, Atul Kedia, Richard O’Shaughnessy, Christopher J. Fontes, Chris L. Fryer, Oleg Korobkin, Matthew R. Mumpower, V. Ashley Villar, and Ryan T. Wollaeger.
Kilonova Light-Curve Interpolation with Neural Networks.
Physical Review Research, 6(3):033078, July 2024.
33. Chris L. Fryer, Aimee L. Hungerford, Ryan T. Wollaeger, Jonah M. Miller, Soumi De, Christopher J. Fontes, Oleg Korobkin, Atul Kedia, Marko Ristic, and Richard O’Shaughnessy.
The Effect of the Velocity Distribution on Kilonova Emission.
Astroph. J., 961(1):9, January 2024.
34. Tousif Islam, Avi Vajpeyi, Feroz H. Shaik, Carl-Johan Haster, Vijay Varma, Scott E. Field, Jacob Lange, Richard O’Shaughnessy, and Rory Smith.
Analysis of GWTC-3 with fully precessing numerical relativity surrogate models.
Phys. Rev. D, 112(4):044001, August 2025.
35. Marko Ristić, Richard O’Shaughnessy, V. Ashley Villar, Ryan T. Wollaeger, Oleg Korobkin, Chris L. Fryer, Christopher J. Fontes, and Atul Kedia.
Interpolated kilonova spectra models: Examining the effects of a phenomenological, blue component in the fitting of AT2017gfo spectra.
Physical Review Research, 5(4):043106, November 2023a.
36. Vera Delfavero, Richard O’Shaughnessy, Krzysztof Belczynski, Paweł Drozda, and Daniel Wysocki.
Iteratively Comparing Gravitational-Wave Observations to the Evolution of Massive Stellar Binaries.
Phys. Rev. D, 108:043023, Aug 2023. (link).
37. A. B. Yelikar, V. Delfavero, and **R. O’Shaughnessy**.
Low-latency parameter inference enabled by a Gaussian likelihood approximation for RIFT.
arXiv e-prints, art. arXiv:2301.01337, January 2023.
38. Atul Kedia, Marko Ristic, Richard O’Shaughnessy, Anjali B. Yelikar, Ryan T. Wollaeger, Oleg Korobkin, Eve A. Chase, Christopher L. Fryer, and Christopher J. Fontes.
Surrogate light curve models for kilonovae with comprehensive wind ejecta outflows and parameter estimation for AT2017gfo.
Physical Review Research, 5(1):013168, March 2023.

39. V. Gayathri, Daniel Wysocki, Y. Yang, Vera Delfavero, **R. O’Shaughnessy**, Z. Haiman, H. Tagawa, and I. Bartos.
Gravitational Wave Source Populations: Disentangling an AGN Component.
Astroph. J. Letters, 945(2):L29, March 2023.
40. J. Wofford, A. B. Yelikar, Hannah Gallagher, E. Champion, D. Wysocki, V. Delfavero, J. Lange, C. Rose, V. Valsan, S. Morisaki, J. Read, C. Henshaw, and **R. O’Shaughnessy**.
Improving performance for gravitational-wave parameter inference with an efficient and highly-parallelized algorithm.
Phys. Rev. D, 107(2):024040, January 2023.
41. H. L. Iglesias, J. Lange, I. Bartos, S. Bhaumik, R. Gamba, V. Gayathri, A. Jan, R. Nowicki, **R. O’Shaughnessy**, D. M. Shoemaker, R. Venkataramanan, and K. Wagner.
Eccentricity Estimation for Five Binary Black Hole Mergers with Higher-order Gravitational-wave Modes.
Astroph. J., 972(1):65, September 2024.
42. Marko Ristić, Erika M. Holmbeck, Ryan T. Wollaeger, Oleg Korobkin, Elizabeth Champion, Richard O’Shaughnessy, Chris L. Fryer, Christopher J. Fontes, Matthew R. Mumpower, and Trevor M. Sprouse.
Constraining Inputs to Realistic Kilonova Simulations through Comparison to Observed r-process Abundances.
Astroph. J., 956(1):64, October 2023b.
43. Karl Daningburg and Richard O’Shaughnessy.
Cost minimization in acquisition for gravitational wave surrogate modeling.
Phys. Rev. D, 106(8):084020, October 2022.
44. V. Delfavero, **R. O’Shaughnessy**, D. Wysocki, and A. Yelikar.
Compressed parametric and non-parametric approximations to the gravitational wave likelihood.
Submitted to PRD, 2022. (link).
45. Chad Henshaw, Richard O’Shaughnessy, and Laura Cadonati.
Implementation of a generalized precession parameter in the RIFT parameter estimation algorithm.
Classical and Quantum Gravity, 39(12):125003, June 2022.
46. Federico G. Lopez Armengol, Zachariah B. Etienne, Scott C. Noble, Bernard J. Kelly, Leonardo R. Werneck, Brendan Drachler, Manuela Campanelli, Federico Cipolletta, Yosef Zlochower, Ariadna Murguia-Berthier, Lorenzo Ennoggi, Mark Avara, Riccardo Ciolfi, Joshua Faber, Grace Fiacco, Bruno Giacomazzo, Tanmayee Gupte, Trung Ha, Julian H. Krolik, Vassilios Mewes, Richard O’Shaughnessy, Jesús M. Rueda-Becerril, and Jeremy Schnittman.
Handing off the outcome of binary neutron star mergers for accurate and long-term postmerger simulations.
Phys. Rev. D, 106(8):083015, October 2022.
47. Erika M. Holmbeck, Richard O’Shaughnessy, Vera Delfavero, and Krzysztof Belczynski.
A Nuclear Equation of State Inferred from Stellar r-process Abundances.
Astroph. J., 926(2):196, February 2022.
48. Vera Delfavero, Richard O’Shaughnessy, Daniel Wysocki, and Anjali Yelikar.
Normal Approximate Likelihoods to Gravitational Wave Events.
arXiv e-prints, art. arXiv:2107.13082, July 2021.
49. B. McKernan, K. E. S. Ford, T. Callister, W. M. Farr, **R. O’Shaughnessy**, R. Smith, E. Thrane, and A. Vajpeyi.
LIGO-Virgo correlations between mass ratio and effective inspiral spin: testing the active galactic nuclei channel.
MNRAS, 514(3):3886–3893, August 2022.
50. Ariadna Murguia-Berthier, Scott C. Noble, Luke F. Roberts, Enrico Ramirez-Ruiz, Leonardo R. Werneck, Michael Kolacki, Zachariah B. Etienne, Mark Avara, Manuela Campanelli, Riccardo Ciolfi, Federico Cipolletta, Brendan Drachler, Lorenzo Ennoggi, Joshua Faber, Grace Fiacco, Bruno Giacomazzo,

- Tanmayee Gupte, Trung Ha, Bernard J. Kelly, Julian H. Krolik, Federico G. Lopez Armengol, Ben Margalit, Tim Moon, Richard O’Shaughnessy, Jesús M. Rueda-Becerril, Jeremy Schnittman, Yossef Zenati, and Yosef Zlochower.
HARM3D+NUC: A New Method for Simulating the Post-merger Phase of Binary Neutron Star Mergers with GRMHD, Tabulated EOS, and Neutrino Leakage.
Astroph. J., 919(2):95, October 2021.
51. R. T. Wollaeger, C. L. Fryer, E. A. Chase, C. J. Fontes, M. Ristic, A. L. Hungerford, O. Korobkin, **R. O’Shaughnessy**, and A. M. Herring.
A Broad Grid of 2D Kilonova Emission Models.
Astroph. J., 918:arXiv:2105.11543, August 2021.
 52. M. Ristic, E. Champion, **R. O’Shaughnessy**, R. Wollaeger, O. Korobkin, E. A. Chase, C. L. Fryer, A. L. Hungerford, and C. J. Fontes.
Interpolating detailed simulations of kilonovae: Adaptive learning and parameter inference applications.
Physical Review Research, 4(1):013046, January 2022. (link).
 53. A. Z. Jan, A. B. Yelkar, J. Lange, and **R. O’Shaughnessy**.
Assessing and marginalizing over compact binary coalescence waveform systematics with RIFT.
Phys. Rev. D, 102(12):124069, December 2020.
 54. Hong Qi, Richard O’Shaughnessy, and Patrick Brady.
Testing the black hole no-hair theorem with Galactic Center stellar orbits.
Phys. Rev. D, 103(8):084006, April 2021.
 55. Mohammad Al-Mamun, Andrew W. Steiner, Joonas Nättilä, Jacob Lange, Richard O’Shaughnessy, Ingo Tews, Stefano Gandolfi, Craig Heinke, and Sophia Han.
Combining Electromagnetic and Gravitational-Wave Constraints on Neutron-Star Masses and Radii.
Phys. Rev. Lett., 126(6):061101, February 2021.
 56. P. Drozda, K. Belczynski, **R. O’Shaughnessy**, T. Bulik, and C. L. Fryer.
Black hole-neutron star mergers: The first mass gap and kilonovae.
A&A, 667:A126, November 2022.
 57. V. Gayathri, J. Healy, J. Lange, B. O’Brien, M. Szczepańczyk, Imre Bartos, M. Campanelli, S. Klimentenko, C. O. Lousto, and **R. O’Shaughnessy**.
Eccentricity estimate for black hole mergers with numerical relativity simulations.
Nature Astronomy, art. arXiv:2009.05461, January 2022. (link).
 58. D Wysocki, **R. O’Shaughnessy**, L. Wade, and J. Lange.
Inferring the neutron star equation of state simultaneously with the population of merging neutron stars.
Submitted to PRD; available as arxiv:2001.01747, January 2020. (link).
 59. K. Belczynski, R. Hirschi, E. A. Kaiser, Jifeng Liu, J. Casares, Youjun Lu, **R. O’Shaughnessy**, A. Heger, and S. Justham.
The Formation of a 70 Msun Black Hole at High Metallicity.
arXiv e-prints, art. arXiv:1911.12357, Nov 2019.
 60. Z. Doctor, D. Wysocki, **R. O’Shaughnessy**, D. E. Holz, and B. Farr.
Black Hole Coagulation: Modeling Hierarchical Mergers in Black Hole Populations.
Astroph. J., 893(1):35, April 2020.
 61. F Shaik, J. Lange, S. Field, **R. O’Shaughnessy**, V Varma, and et al.
Impact of subdominant modes on the interpretation of gravitational-wave signals from heavy binary black hole systems.
Phys. Rev. D, 101:124054, July 2020. (link).
 62. Y. Yang, I. Bartos, V. Gayathri, S. Ford, Z. Haiman, S. Klimentenko, B. Kocsis, S. Márka, Z. Márka, B. McKernan, and R. O’Shaughnessy.
Hierarchical Black Hole Mergers in Active Galactic Nuclei.

- Phys. Rev. Lett.*, 123:181101, November 2019.
URL (link).
63. B. McKernan, K. E. S. Ford, R. O’Shaughnessy, and D. Wysocki.
Monte Carlo simulations of black hole mergers in AGN discs: Low χ_{eff} mergers and predictions for LIGO.
MNRAS, 494(1):1203–1216, May 2020.
 64. K. Belczynski, T. Bulik, A. Olejak, M. Chruslinska, N. Singh, N. Pol, L. Zdunik, **R. O’Shaughnessy**, M. McLaughlin, D. Lorimer, O. Korobkin, E. P. J. van den Heuvel, M. B. Davies, and D. E. Holz.
Binary neutron star formation and the origin of GW170817.
Available as arxiv:1812.10065, December 2018.
 65. D. Wysocki, J. Lange, and **R. O’Shaughnessy**.
Reconstructing phenomenological distributions of compact binaries via gravitational wave observations.
Phys. Rev. D, 100:3012, August 2019. (link).
 66. J. Lange, **R. O’Shaughnessy**, and M. Rizzo.
Rapid and accurate parameter inference for coalescing, precessing compact binaries.
Submitted to PRD; available at arxiv:1805.10457, 2018.
 67. M. W. Coughlin, T. Dietrich, Z. Doctor, D. Kasen, S. Coughlin, A. Jerkstrand, G. Leloudas, O. McBrien, B. D. Metzger, **R. O’Shaughnessy**, and S. J. Smartt.
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1. Reducing Thermoelastic Noise by Reshaping the Light Beams and Test Masses. *LIGO Scientific Collaboration general meeting*, August 2001.
2. Reducing thermoelastic noise in LIGO mirrors. *17th Pacific Coast Gravity Meeting, ITP, UC Santa Barbara*, March 2001.
3. Mexican Hat (Flat-Topped) Beams for Advanced LIGO. *LIGO Scientific Collaboration general meeting*, August 2003.
4. Constraints on binary black hole inspiral rates via population synthesis and binary neutron stars. *17th International Conference on General Relativity and Gravitation*, December 2004.
5. Population synthesis and binary black hole merger rates. *LIGO Scientific Collaboration general meeting*, August 2004.
6. Population synthesis and binary black hole merger rates. *Midwest Relativity Meeting*, October 2004.
7. Constraints on compact-object merger rates via (EM) NS-NS observations. *9th annual Gravitational Wave Data Analysis Workshop*, December 2004.
8. Phase steps: Nonparametric extensions to inspiral template families. *9th annual Gravitational Wave Data Analysis Workshop*, December 2004.

9. Expected compact-object merger rates. *LIGO Scientific Collaboration general meeting*, March 2005.
10. Black hole mergers via interactions in dense clusters (**Invited talk**). *MODEST-6 Meeting, Northwestern University*, August 2005.
11. Delayed mergers: The contribution of ellipticals, globular clusters, and protoclusters to the LIGO detection rate. *LIGO Scientific Collaboration general meeting*, August 2005.
12. Updated merger rates BH-BH, BH-NS, NS-NS rates via best-constrained population synthesis. *LIGO Scientific Collaboration general meeting*, August 2005.
13. Binary models for short gamma ray bursts (**Invited talk**). *New Views of the Universe, KICP Inaugural Symposium, Kavli Institute, Chicago*, December 2005.
14. Compact object merger rates. *10th annual Gravitational Wave Data Analysis Workshop*, December 2005.
15. Critically assessing Binary mergers as short hard GRBs (**Invited talk**) . *LIGO-Caltech*, March 2006.
16. Compact object merger rates: Predictions and Constraints (including short GRBs). *LIGO Scientific Collaboration general meeting*, March 2006.
17. Detecting binary mergers with gravitational waves (or, *Why LIGO is needed to understand short GRBs*) (**Invited talk**). *Argonne National Lab, High Energy Physics/Astrophysics seminar*, November 2006.
18. Constraining binary evolution with event rates: Predicted merger rates of and astrophysical constraints on BH-NS and NS-NS mergers, including short GRBs. *23rd Texas Symposium, Melbourne, Australia*, December 2006.
19. Gravitational wave astronomy: A new window on the universe (**Invited talk**). *Georgia Tech Physics Colloquium, Atlanta, GA*, January 2007.
20. Astrophysical constraints on BH-NS and NS-NS mergers and the short GRB redshift distribution (**Invited talk**). *KICP Lunch Colloquium, Chicago, IL*, February 2007.
21. Astrophysical constraints on BH-NS and NS-NS mergers and the short GRB redshift distribution (**Invited talk**). *UIUC Physics Colloquium, Champaign, IL*, 2007.
22. Comparing the known (astrophysical constraints on BH-NS and NS-NS mergers) and the unknown (short GRBs) (**Invited talk**). *Ringberg short GRB workshop, Munich, Germany*, 2007.
23. Unravelling short GRBs with LIGO, Swift, and GLAST. *Argonne GLAST workshope, Argonne National Lab, Chicago, USA*, 2007.
24. Short GRBs and Mergers: Astrophysical constraints on a BH-NS and NS-NS origin. *APS, Jacksonville, USA*, 2007.
25. Short GRBs and Mergers: Astrophysical constraints on a BH-NS and NS-NS origin. *Penn State University, USA*, May 2007.
26. The probability of compact binary coalescence detection with enhanced LIGO. *Boston, MA*, December 2007.
27. Can short GRBs be NS mergers? *APS April meeting*, April 2008.
28. Astrophysics with LIGO (**Invited talk**). *Columbia University Particle Physics Seminar*, 2008.
29. Astrophysics with LIGO: Constraining binary populations. *East Coast Gravity Meeting*, May 2008.
30. Ground-based gravitational-wave astronomy and compact objects in clusters (**Invited talk**). *UCSB KITP globular cluster program*, 2009.
31. Astrophysics with gravitational-wave measurements of binary compact object mass distributions. *APS April Meeting*, May 2009.
32. Gravitational wave emission from (short) GRBs (**Invited talk**). *Multi-messenger astrophysics conference: Center for relativistic astrophysics, Georgia Tech*, May 2009.
33. Gravitational wave signatures of binary evolution (**Invited talk**) . *Syracuse university gravitational wave group*, May 2009.

34. Astrophysics with gravitational-wave measurements of binary compact object mass distributions. *8th Amaldi meeting*, May 2009.
35. Binary neutron star astrophysics (**Invited talk**). *Numerical relativity and data analysis conference; AEI, Potsdam*, 2009.
36. Opening a new window on the universe: Gravitational wave astronomy with compact binaries (**Invited talk**). *TAMU Commerce, Commerce, TX*, 2010.
37. Using short GRBs to limit birefringence in Chern-Simons modified gravity. *Workshop on gravitational wave tests of alternative theories of gravity in the advanced detector era; UWM, Milwaukee*, May 2010.
38. Testing GR using Externally Triggered Searches: Astrophysical challenges. *Workshop on gravitational wave tests of alternative theories of gravity in the advanced detector era; UWM, Milwaukee*, May 2010.
39. Selection biases of nonspinning searches for spinning binaries in ground-based detector data. *Midwest relativity meeting; Guelph, Canada*, November 2010.
40. Choosing precessing black hole binary simulations that cover the waveform space. *APS April Meeting; Anaheim, CA*, 2011.
41. Microphysics to macrophysics: Astrophysics and gravitational wave science targets in the advanced detector era and beyond. *Microphysics in Computational Relativistic Astrophysics: 2011; Perimeter Institute*, 2011.
42. Low metallicity star formation: a nursery for compact binary mergers? (**Invited talks**). *Perimeter and CITA, Toronto*, October 2011.
43. Low metallicity star formation: a nursery for compact binary mergers? (**Invited talk**). *Center for Relativistic Astrophysics, Georgia Tech*, February 2012.
44. Distinguishing between merging black hole binaries, with orientation-dependent emission. *APS April Meeting; Atlanta, GA*, April 2012.
45. Gravitational wave source populations (**Invited talk**). *Connecting the Gravitational Wave and Electromagnetic Skies in the Era of Advanced LIGO, Princeton, NJ*, May 2012.
46. Precession during merger: Strong polarization changes are observationally accessible features of strong-field gravity during binary black hole merger. *Rattle and Shine: Gravitational Wave and Electromagnetic Studies of Compact Binary Mergers, KITP, Santa Barbara [July]; and Midwest Relativity Meeting, Chicago, IL [Sept]*, July 2012.
47. Gravitational wave astronomy of merging compact binaries (**Invited talk**). *Workshop on Outstanding Problems in Massive Star Research – the final stages; St. Paul, MN*, October 2012.
48. Interpreting the complex gravitational wave symphony from merging, precessing black hole binaries. *Physics colloquium, Department of Physics, U. of Mississippi (Invited talk)*, November 2012.
49. Low metallicity star formation: a nursery for compact binary mergers? *Theoretical Astrophysics and Relativity Seminar, Caltech*, January 2013.
50. Gravitational wave astrophysics with merging compact binaries (**Invited talk**). *Theoretical astrophysics seminar, U. Illinois (Urbana)*, February 2013.
51. Interpreting the complex gravitational wave symphony from merging, precessing black hole binaries (**Invited talk**). *Princeton [March] and RIT [June]*, March 2013.
52. Using a corotating frame to model and interpret gravitational waves from strong-field binary black hole merger. *APS Meeting, Denver, CO*, April 2013.
53. Gravitational wave source populations. *Caltech Gravitational Wave Astrophysics summer school*, July 2013.
54. A closed-form model for gravitational waves from precessing BH-NS binaries. *23rd Midwest Relativity Meeting, Milwaukee, WI*, October 2013.
55. Disentangling astrophysics from gravity. *Testing General Relativity with Astrophysical Observations, Oxford MS*, January 2014.

56. Astrophysical inferences from gravitational wave detections. *SMS colloquium, Rochester Institute of Technology [February]; Physics colloquium, U. Mississippi [March]*, 2014.
57. Estimating parameters of BH-NS binaries with gravitational waves. *APS April meeting, Savanna, GA*, April 2014.
58. Efficient high-mass parameter estimation. *NARDA 2014, Fullerton, CA*, August 2014.
59. Understanding and evolving precessing black hole binaries. *Aspen Center for Physics*, January 2015.
60. A semianalytic Fisher matrix for precessing BH-NS binaries. *APS April Meeting, Baltimore, MD*, April 2015.
61. Efficient high-mass parameter estimation. *Cornell University relativity seminar, Ithaca, NY*, May 2015.
62. Efficient high-mass parameter estimation. *2015 East Coast Gravity Meeting, Rochester, NY*, May 2015.
63. New perspectives on the dynamics of precessing binary black holes. *CITA (U. Toronto)*, October 2015.
64. GR@100++: Beyond Gravitational Wave Detection (invited panelist). *Princeton University, Center for Theoretical Science*, April 2016.
65. Directly comparing GW150914 with Numerical Relativity. *APS April Meeting, Salt Lake City*, April 2016.
66. Directly comparing GW150914 with Numerical Relativity. *Gravitational Wave Physics and Astronomy Workshop*, July 2016.
67. Will LIGO infer formation scenarios of binary BH mergers (**Invited talk**). *KITP Rapid Response Workshop: Astrophysics from LIGO's First Black Holes*, August 2016.
68. Data analysis for the future (Roundtable). *GW161212: The Universe through gravitational waves, Simons Center for Geometry and Physics, Stony Brook NY*, December 2016.
69. An architecture for efficient multimodal parameter estimation with linear surrogate models. *APS April Meeting, Washington DC*, January 2017.
70. Numerical relativity and gw data. *The Dawning Era of Gravitational-Wave Astrophysics, Aspen Center for Physics*, January 2017.
71. Measuring the imprint of spin in the strong field. *Strong Gravity and Binary Dynamics*, February 2017.
72. Inferences about supernova physics from gravitational-wave measurements of gw151226. June 2017.
73. How much does galaxy assembly history impact the population of merging compact binaries? (**Invited talk**). *Texas Tech University*, June 2017.
74. Forming ligo's merging bh-bh binaries via isolated binary evolution. *Aspen Center for Physics*, July 2017.
75. Gravitational waves: opening a new window on the universe. November 2017.
76. Compact object astrophysics and gravitational wave astronomy (**invited talk**). *NY section APS meeting*, November 2017.
77. Inferring tidal distortion of coalescing neutron star binaries. *231st American Astronomical Society Meeting, Washington DC*, January 2018.
78. Constraining the nuclear equation of state with multiple observational channels. *April APS meeting, Columbus OH*, April 2018.
79. Illuminating a dark world: Gravitational wave astrophysics with binary black holes. *Physics colloquium, U. Tennessee-Knoxville*, 2018.
80. The observed population of compact mergers. *Aspen Center for Physics, GWPop 2019*, February 2019.

81. Primer on Gravitational Wave Observables (**Invited talk**). *Workshop on Stellar Mass black hole mergers in AGN disks, Center for Computational Astrophysics, Flatiron Institute, NY*, March 2019.
82. Astrophysics with gravitational wave populations (**Invited talk**). *Columbia University, NYC, NY*, April 2019.
83. Gravitational Wave Parameter and Population Inference. *Inference in Multimessenger Astrophysics Workshop, UC Berkeley, CA*, May 2019.
84. Update on LIGO Astrophysics. *Advancing Computational Methods to understand the dynamics of ejection, Accretion, Winds and Jets in Neutron Star Mergers workshop, RIT, Rochester, NY*, 2019.
85. Dynamical environments are interesting iii. *KITP Program: The New Era of Gravitational-Wave Physics and Astrophysics, Santa Barbara, CA*, July 2019.
86. Simultaneous inference of the neutron star population and equation of state. *APS April Meeting*, 2020.
87. What can we learn about NS in binaries with gravitational waves? *Advancing Computational Methods to understand the dynamics of ejection, Accretion, Winds and Jets in Neutron Star Mergers workshop, RIT, Rochester, NY*, July 2020.
88. LIGO O3: the story so far (**Invited talk**). *AGN Disks: Where the Wild Things Are, Center for Computational Astrophysics, Flatiron Institute, NY*, October 2020.
89. Multimessenger and multiobject parameter inference. *The r-process and the nuclear EOS after LIGO-Virgo's third observing run, Institute for Nuclear Theory, University of Washington, Seattle, WA*, 2022.
90. Iteratively interpreting Gravitational wave sources and populations. *Physics colloquium, CSU Fullerton*, August 2022.
91. Gravitational wave sources 1: The story so far (**Invited talk**). *Miami 2023 Physics Conference*, December 2023.
92. Learning from gravitational waves (**Invited talk**). *"Multi-messenger Transients from Binary Mergers and Stellar Explosions", Aspen center for physics, Aug 4-24 2024*, August 2024.

AWARDS AND HONORS

Bruno Rossi Prize (2017, with Gaby Gonzalez as part of the LIGO Scientific Collaboration)
 Princess of Asturias Award for Scientific and Technical Research (2017, as part of the LIGO Scientific Collaboration)
 Einstein Medal (2017, as part of the LIGO Scientific Collaboration)
 Special Breakthrough Prize in Fundamental Physics (2016, as part of the LIGO Scientific Collaboration)
 2016 Gruber Cosmology Prize (as part of the LIGO Scientific Collaboration)